Tutorial 16 — 24/04/2016

1. $f(z,t) = A \sin(kz)\cos(kvt)$

$$= \frac{A}{2}\Big[\underbrace{\sin k(z+vt)}_{\substack{\text{represents wave}\\ \text{traveling in -ve}\\ \text{z-axis}}} + \underbrace{\sin k(z-vt)}_{\substack{\text{represents wave}\\ \text{traveling in +ve}\\ \text{z-axis.}}}\Big]$$

$$\frac{\partial f}{\partial z} = kA\cos(kz)\cos(kvt)$$

$$\frac{\partial^2 f}{\partial z^2} = -k^2 A\sin(kz)\cos(kvt)$$

$$= -k^2 f(z,t)$$

$$\frac{\partial f}{\partial t} = -kv\,A\sin(kz)\sin(kvt)$$

$$\frac{\partial^2 f}{\partial t^2} = -k^2v^2 A\sin(kz)\cos(kvt)$$

$$= -k^2v^2 f(z,t)$$

$$\boxed{\frac{\partial^2 f}{\partial z^2} = \frac{1}{v^2}\frac{\partial^2 f}{\partial t^2}}$$

$f(z,t)$ satisfies classical wave equation

2. We have seen that plane monochromatic waves of $\vec{E}$ and $\vec{B}$ can be represented in the form (in vacuum)

$$\vec{E}(\vec{r},t) = E_0\cos(\vec{k}\cdot\vec{r} - \omega t + \delta)\,\hat{n} \quad \text{(i)}$$ [$\hat{n}$: direction of polarization of $\vec{E}$ vectors]

$$\vec{B}(\vec{r},t) = \frac{E_0}{c}\cos(\vec{k}\cdot\vec{r} - \omega t + \delta)(\hat{k}\times\hat{n}) \quad \text{(ii)}$$

Here $\hat{k}$ denotes direction of propagation of electromagnetic wave.
δ = phase delay (zero in the question)

(a) Here e.m wave is traveling in -ve x direction. The direction of polarization of e.m wave (z-axis) typically refers to the direction of vibration of electric field vectors. As this is a one-dimensional problem, we may write
$\vec{E}(x,t) = E_0\cos(kx + \omega t)\,\hat{z}$. This is also clear from eq(i)

$\vec{k} = \frac{\omega}{c}\hat{k} = \frac{\omega}{c}(-\hat{x})$ ($\hat{k} = -\hat{x}$ in question)

(i) reduces to

$$\boxed{\vec{k}\cdot\vec{r} = \left(-\frac{\omega}{c}\right)\hat{x}\cdot(\hat{x}x + \hat{y}y + \hat{z}z) = -\frac{\omega}{c}x}$$

$$\vec{E}(x,t) = E_0 \cos\left(-\frac{\omega}{c}x - \omega t\right)\hat{z}$$

$$= E_0 \cos\left(\omega\left(\frac{x}{c}\right) + \omega t\right)\hat{z}$$

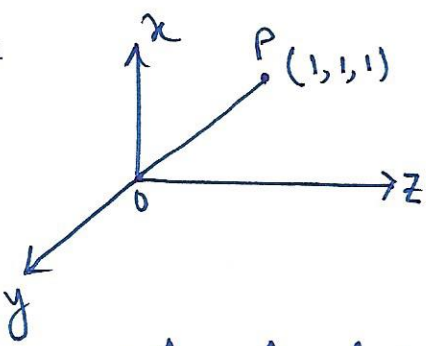
$$\vec{E}(\vec{x},t) = E_0 \cos(kx + \omega t) \quad \left[k = |\vec{k}| = \frac{\omega}{c}\right]$$

(ii) reduces to

$$\vec{B}(x,t) = \frac{E_0}{c}\cos\left(-\frac{\omega}{c}x - \omega t\right)(-\hat{x}\times\hat{z})$$

$$= \frac{E_0}{c}\cos(kx + \omega t)\,\hat{y}$$

(b) Now the wave is traveling in the direction of OP



$\vec{k} = \left(\frac{\omega}{c}\right)\hat{OP}$ [$\hat{OP}$ = unit vector along OP]

$$\hat{OP} = \frac{(1-0)\hat{x} + (1-0)\hat{y} + (1-0)\hat{z}}{\sqrt{1^2+1^2+1^2}}$$

$$\boxed{= \frac{1}{\sqrt{3}}\hat{x} + \frac{1}{\sqrt{3}}\hat{y} + \frac{1}{\sqrt{3}}\hat{z}\ (= \hat{k})}$$

$$\vec{k} = \frac{\omega}{c}\left(\frac{\hat{x}}{\sqrt{3}} + \frac{\hat{y}}{\sqrt{3}} + \frac{\hat{z}}{\sqrt{3}}\right)$$

Also e.m wave is polarized along x-z plane

$$\hat{n} = \alpha\hat{x} + \beta\hat{z}$$

Since we know that monochrometic plane e.m waves are transverse

$\hat{n}\cdot\hat{k} = 0 \Rightarrow \frac{\alpha}{\sqrt{3}} + \frac{\beta}{\sqrt{3}} = 0 \Rightarrow \alpha = -\beta$. Thus $\hat{n} = \alpha(\hat{x} - \hat{z})$. Also

Since $|\hat{n}| = 1 \Rightarrow \alpha = \frac{1}{\sqrt{2}}$. Thus $\boxed{\hat{n} = \frac{\hat{x}}{\sqrt{2}} - \frac{\hat{z}}{\sqrt{2}}}$

$$\vec{k}\cdot\vec{r} = \frac{\omega}{c}\left(\frac{\hat{x}}{\sqrt{3}} + \frac{\hat{y}}{\sqrt{3}} + \frac{\hat{z}}{\sqrt{3}}\right)\cdot(\hat{x}x + \hat{y}y + \hat{z}z)$$

$$\vec{k}\cdot\vec{r} = \frac{\omega}{c}\frac{1}{\sqrt{3}}(x+y+z)$$

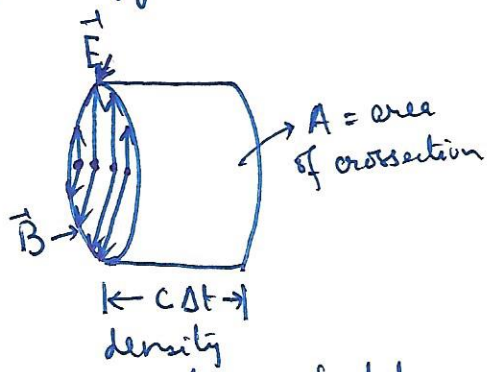
$$\hat{k}\times\hat{n} = \frac{1}{\sqrt{3}}\cdot\frac{1}{\sqrt{2}}\begin{vmatrix}\hat{x} & \hat{y} & \hat{z}\\ 1 & 1 & 1\\ 1 & 0 & -1\end{vmatrix} = \frac{1}{\sqrt{6}}(-\hat{x}+2\hat{y}+\hat{z})$$

$$\vec{E}(r,t) = E_0\cos\left[\frac{\omega}{\sqrt{3}c}(x+y+z)-\omega t\right]\left(\frac{\hat{x}-\hat{z}}{\sqrt{2}}\right)$$

$$\vec{B}(r,t) = \frac{E_0}{c}\cos\left[\frac{\omega}{\sqrt{3}c}(x+y+z)-\omega t\right]\left(\frac{-\hat{x}+2\hat{y}-\hat{z}}{\sqrt{6}}\right)$$

3. Intensity of sunlight = 1300 W/m^2

Pressure exerted by electromagnetic waves on a perfect absorber can be calculated in the following way (see slides) – electromagnetic waves of length $c\Delta t$, in ~~vacuum~~ vacuum crosses area A carrying energy $uAc\Delta t$ where u = energy/volume (energy density)



Now change in momentum when e.m waves hits a perfect absorber is given by

$$\Delta p = \underline{\langle g\rangle Ac\Delta t}, \quad \text{—(iii)}$$

where $\langle g\rangle$ is the time-averaged momentum density of e.m fields

It has been shown that $\langle g\rangle = \frac{1}{2c}\epsilon_0 E_0^2$ is the time averaged momentum density (momentum per unit volume) of e.m fields.

Radiation pressure, $\frac{\Delta p}{\Delta t}\times\frac{1}{A} = \langle g\rangle c = \frac{1}{2}\epsilon_0 E_0^2$. Also intensity of e.m radiation in the form of plane waves is given by

$$I = \langle S \rangle = \frac{1}{2} c \epsilon_0 E_0^2$$ (E_0 = amplitude of electric field)

↑ time averaged Poynting vector.

(The proofs have been explained in class)

Thus radiation pressure $\boxed{P = \frac{I}{c}}$

For perfect reflectors, from eqn (iii) momentum switches its direction, instead of simply being absorbed – so pressure is twice as great.

Coming back to the problem

$$P = \frac{1300\ W/m^2}{3\times10^8\ m/s} = 4.3\times10^{-6}\ N/m^2$$

As discussed, for perfect reflector, $P = 8.6\times10^{-6}\ N/m^2$

$1\ atm = 1.03\times10^5\ N/m^2$

Pressure of light on reflector $= \frac{8.6\times10^{-6}}{1.03\times10^{-5}} = 8.3\times10^{-11}$ atm.

4. Here $b = 8.8\times10^{-3}$ cm (slit width of each slit)

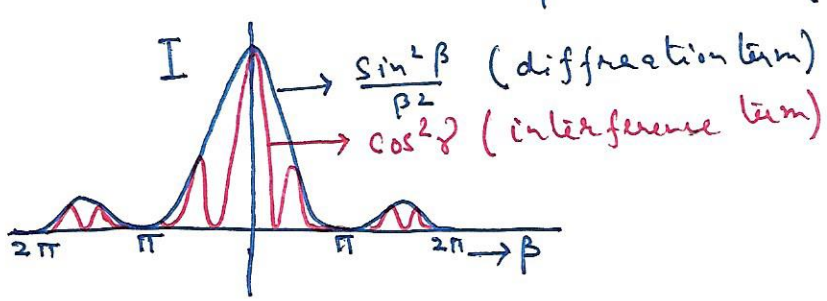
$d = 7.0\times10^{-2}$ cm (distance between 2 slits)

$\lambda = 6.328\times10^{-5}$ cm (excitation wavelength)

Intensity distribution in 2-slit diffraction

$$I = 4 I_0 \frac{\sin^2\beta}{\beta^2} \cos^2\gamma \quad \text{——(V)} \qquad \beta = \frac{\pi b \sin\theta}{\lambda}$$

$$\gamma = \frac{\pi d \sin\theta}{\lambda}$$



Diffraction minimum occurs at $\beta = m\pi$ ($m \neq 0$, because $\beta = 0$ corresponds to maxima)
(from eqn. 8(v))

or, $b \sin\theta = m\lambda$

$$\sin\theta = \frac{m\lambda}{b} \quad (m = 1, 2, 3, \ldots)$$

On either side of principal maxima ($\beta = 0$), diffraction minima occurs at

$$\sin\theta = \pm \frac{\lambda}{b} = \pm \frac{6.328 \times 10^{-5}}{8.8 \times 10^{-3}} = 7.2 \times 10^{-3}$$

Interference minima occurs at $\delta = (n + \frac{1}{2})\pi$ (eqn. v)

or, $d \sin\theta = (n + \frac{1}{2})\lambda \quad (n = 0, 1, 2, \ldots)$

$$\sin\theta = (n + \tfrac{1}{2})\frac{\lambda}{d}$$

$$\frac{\lambda}{d} = \frac{6.328 \times 10^{-5}}{7 \times 10^{-2}} \simeq 9 \times 10^{-4} \text{ radians}$$

n	$\sin\theta$ (radians)
0	4.5×10^{-4}
1	$\frac{3}{2} \times 9 \times 10^{-4} = 1.35 \times 10^{-3}$
2	2.26×10^{-3}
3	3.16×10^{-3}
4	4.0×10^{-3}
5	4.97×10^{-3}
6	5.87×10^{-3}
7	6.78×10^{-3}
8	7.68×10^{-3}

Thus there would be 8 ($n = 0$, to $n = 7$) interference minima between two diffraction minima.

Angular seperation between 2 interference maxima,

$\delta = n\pi$

or, $d \sin\theta = n\lambda$

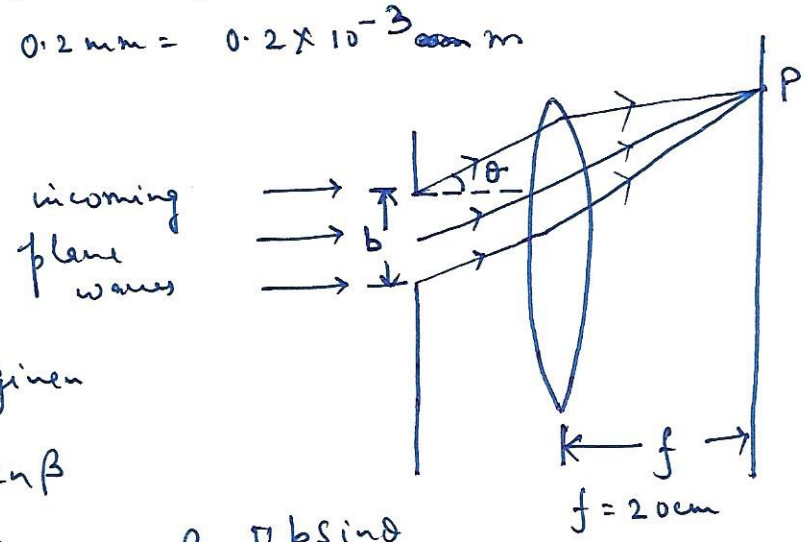
$n = 0, 1, 2, \ldots$ (eqn v)

$$\Delta\theta \approx \frac{\lambda}{d} = 9 \times 10^{-4} \text{ rad}$$

fringe width $= 15 \times 12 \times 2.54 \times 9 \times 10^{-4}$ cm

$= 0.413$ cm

5. This a problem of a single slit diffraction. Width of the slit, $b = 0.2\,mm = 0.2 \times 10^{-3}\,m$



Intensity distribution

$$I = 4I_0 \frac{\sin^2\beta}{\beta^2}$$

Maxima $\left(\frac{dI}{d\beta} = 0\right)$ is given by the equation $\beta = \tan\beta$

$$\beta = \frac{\pi b \sin\theta}{\lambda}$$

Solving graphically, this gives $\beta = 1.43\pi$ and $\beta = 2.46\pi$ as the <u>first two maxima</u>

$b\sin\theta = 1.43\lambda$

$$\sin\theta = \frac{1.43 \times 5\times10^{-5}}{0.02}$$

$\sin\theta \simeq 3.57 \times 10^{-3}$ radians.

$b\sin\theta = 2.46\lambda$

$$\sin\theta = \frac{2.46 \times 5\times10^{-5}}{0.02} = 5\times10^{-3}\,rad$$

$\sin\theta \simeq \theta = 6.15 \times 10^{-3}$ rad.

Since this is small, we may write, $\sin\theta \simeq \theta = 3.5\times10^{-3}$

Angular separation between two maxima $(6.15 - 3.57) \times 10^{-3}$ rad $\simeq 2.5\times10^{-3}$ rad

Distance of separation between two maxima

$= 2.5\times10^{-3} \times f = 2.5\times10^{-3} \times 20$

$= \underline{0.05\,cm}$

Minima is given by $\beta = m\pi$ $(m \neq 0)$

$b\sin\theta = m\lambda \Rightarrow \sin\theta = \frac{\lambda}{b}, \frac{2\lambda}{b}$

$$\sin\theta = \frac{5\times10^{-5}}{0.02} = 2.5\times10^{-3}$$

$\sin\theta \simeq \theta = 2.5\times10^{-3}$ rad.

[$\because \theta$ = small]

$$\sin\theta = \frac{2 \times 5\times10^{-5}}{0.02} = 5\times10^{-3}$$

$\sin\theta \simeq \theta = 5\times10^{-3}$

[$\because \theta$ is small]

Angular separation between first two minima

$= (5 - 2.5) \times 10^{-3}$ rad

$= 2.5 \times 10^{-3}$ rad

Distance of separation = angular separation × focal length

$= 2.5 \times 10^{-3} \times 20 = \underline{0.05 \text{ cm}}$